

MANUFACTURING ANALYTICS





Presented By :

Team Members (Group - 6) :

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PROJECT SUMMARY

Objectives:

- Analyze manufacturing data to monitor production **efficiency and quality**.
- Identify key performance indicators (KPIs) such as **production volume, rejection %, and wastage trends**.
- Develop dashboards in **Excel, Tableau, and Power BI** for effective decision-making.
- Use **SQL queries** for structured data transformation and validation.

Impact:

- Improved visibility into **department-wise performance**.
- Early detection of **high rejection areas, reducing wastage and costs**.
- Data-driven **decision-making** for process optimization.
- Enhanced productivity and decision-making through **interactive dashboards**.

Key Performance Indicators:

1. Manufacture Qty:

Total quantity of items produced within a given timeframe.



2. Rejected Qty:

Number of units that failed quality checks and were rejected.



3. Processed Qty:

Quantity of items that completed specific manufacturing stages.



4. Total Wastage Qty %:

Materials or items lost during the production process.



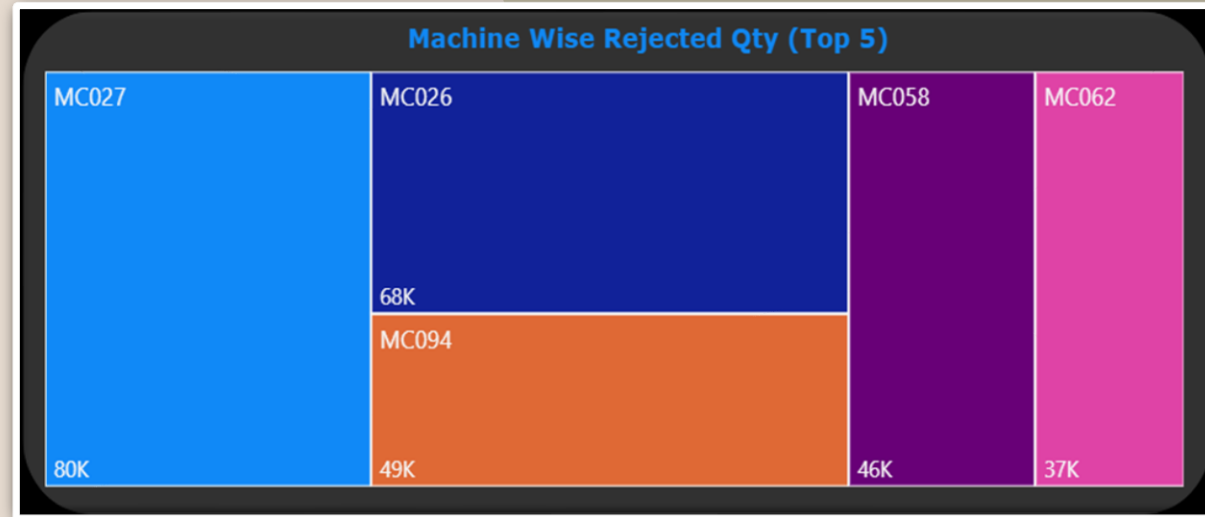
5. Employee Wise Rejected Qty:

Rejection count attributed to individual employees for accountability and training.

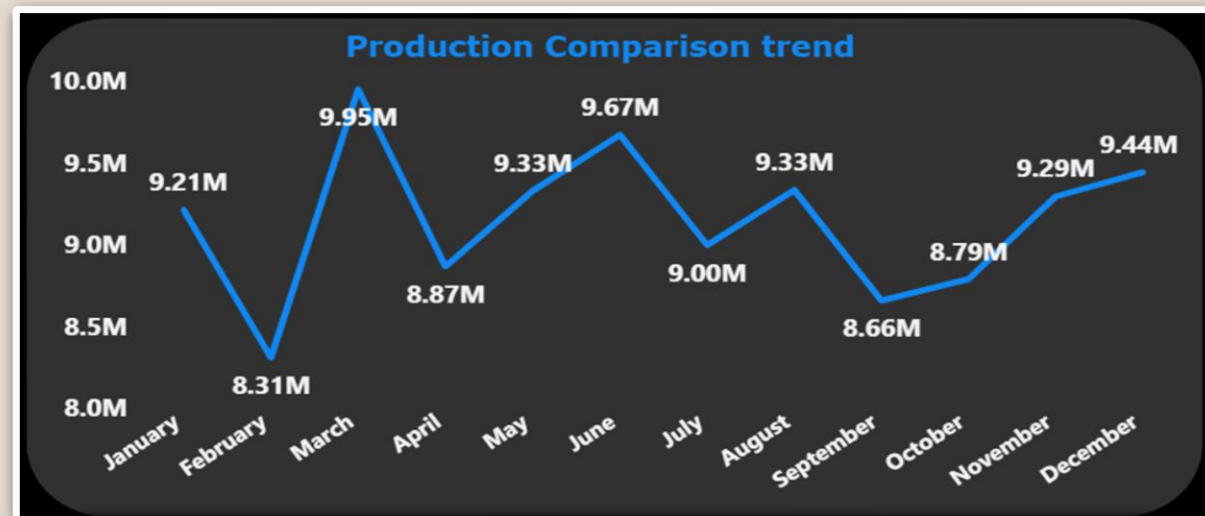


Key Performance Indicators:

6. **Machine Wise Rejected Qty (TOP 5):**
Tracks machine-specific rejections to identify underperforming equipment.



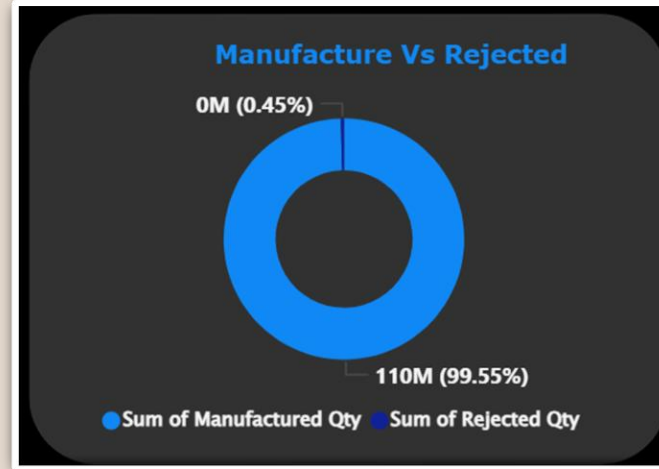
7. **Production Comparison Trend:**
Visual trends of production over time to analyze performance patterns.



Key Performance Indicators:

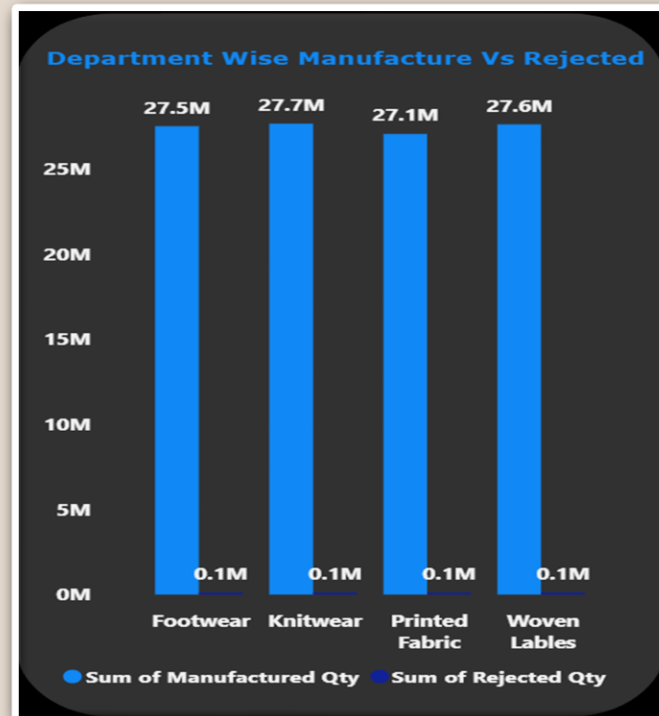
8. Manufacture Vs Rejected:

Direct comparison of produced and rejected quantities to assess quality ratios.



9. Department Wise Manufacture Vs Rejected:

Department-level breakdown of production versus rejection to identify area-specific issues.





EXCEL DASHBOARD

Manufacturing Dashboard

Sales

1,003 M

Manufactured

49790600

Rejected

491023

Buyer



H&M

Nike

Uniqlo

Zara

Departme...

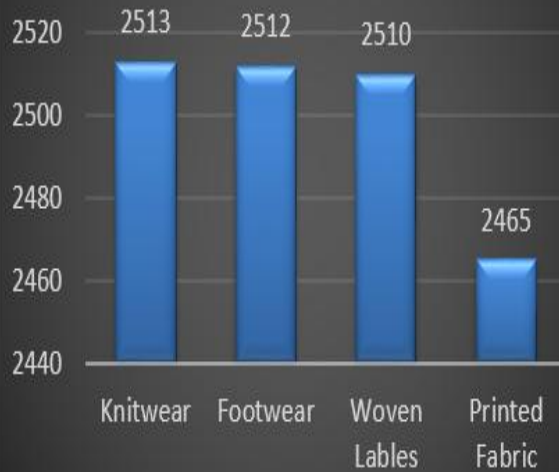


Footwear

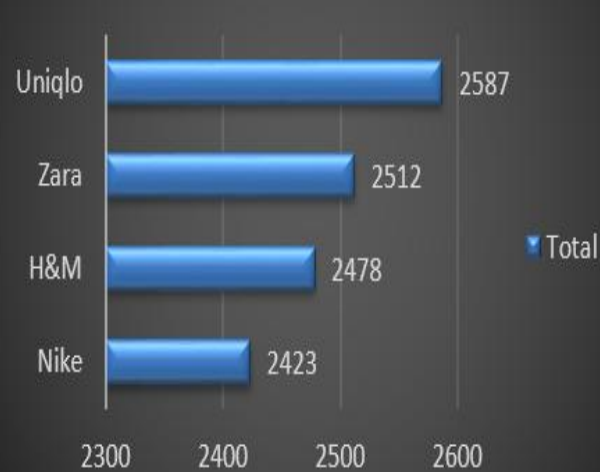
Knitwear

Printed Fabric

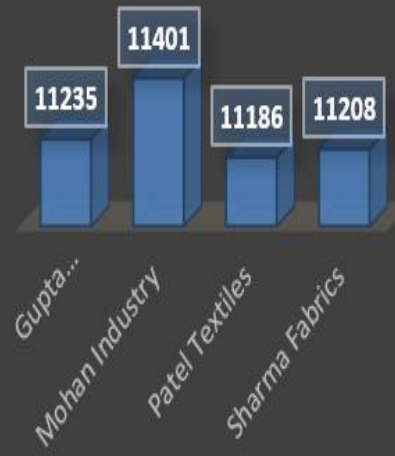
Orders by Dept.



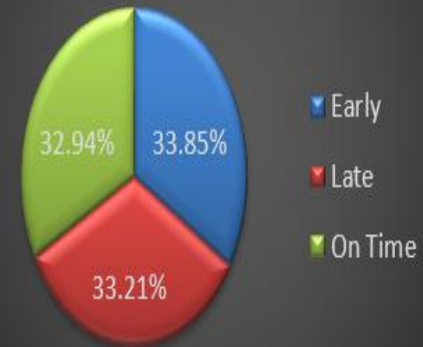
Orders by Buyers



CUSTOMER LOYALTY



Delivery Period



Avg Values By Month



Manufactured Qty Vs WO Qty





TABLEAU DASHBOARD

Manufacturing Dashboard

Total Revenue

1,002.90M

Manufactured Qty

49.79M

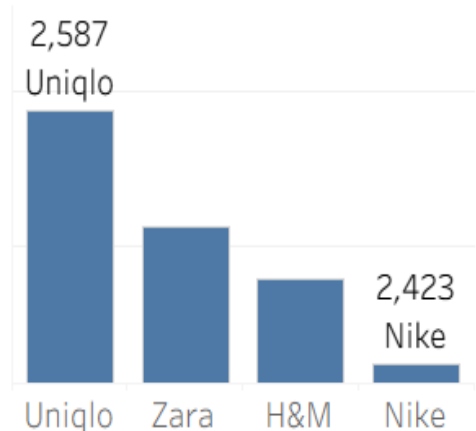
Rejected Qty

0.49M

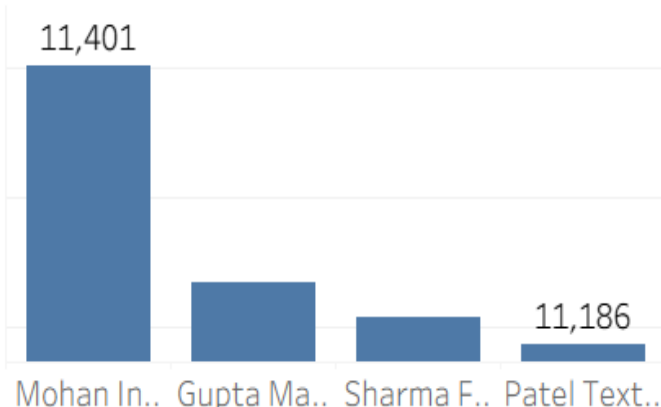
Total Wastage %

0.9960

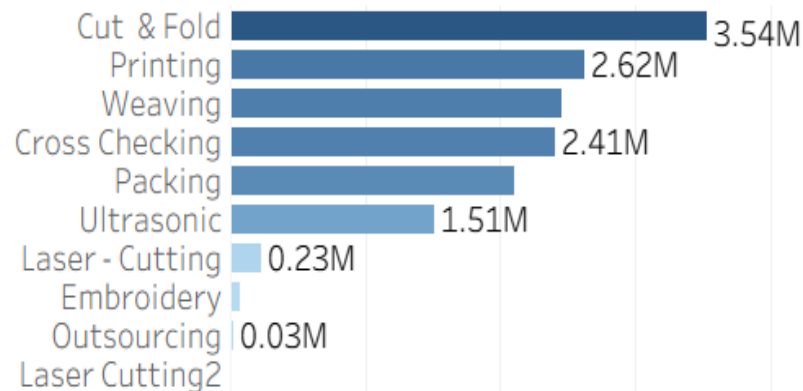
Top Buyers



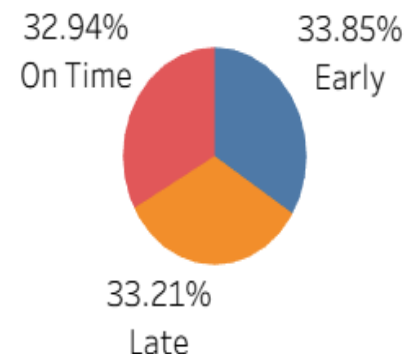
Repeated Customers



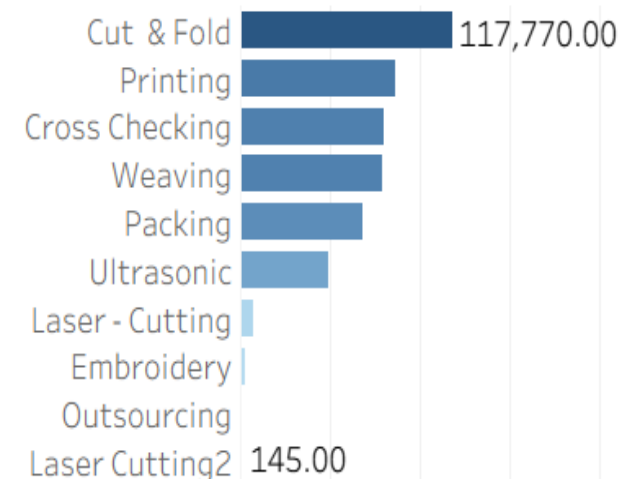
Operation wise Machine cost



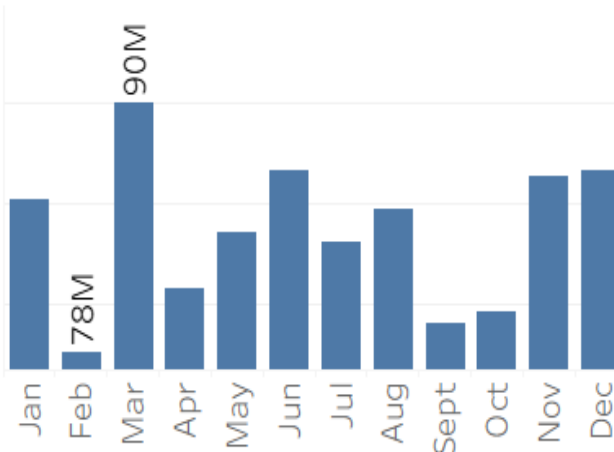
Delivery Period



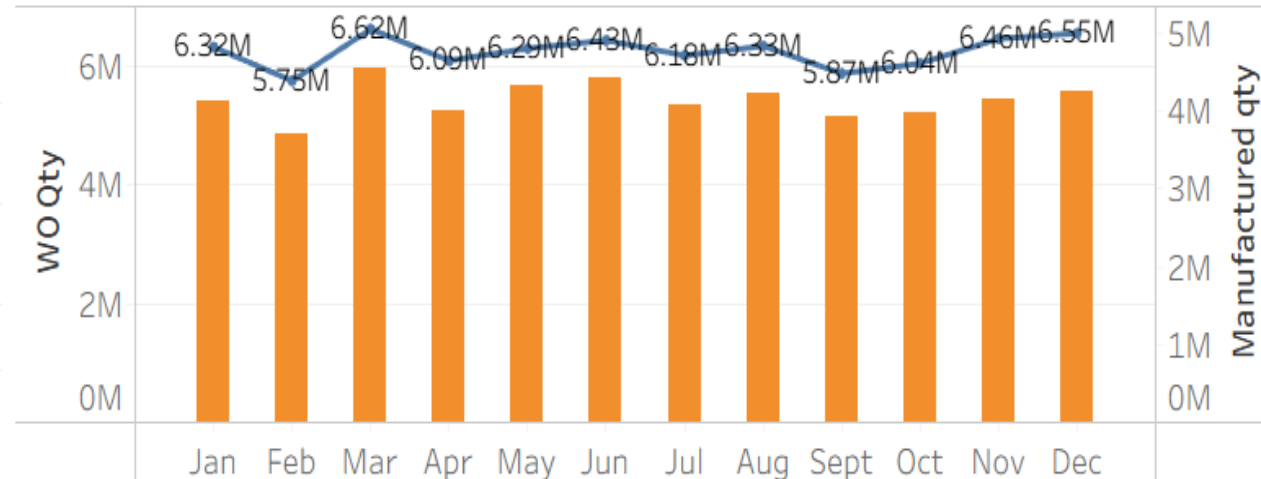
Operation wise rejected qty



Month wise average of revenue



Target Vs Achieved





POWER BI DASHBOARD



Manufacturing Dashboard

Emp Wise

Operation Wise

Machine Wise



Emp Name

All

Department ...

- ☐ Footwear
- ☐ Knitwear
- ☐ Printed Fabric
- ☐ Woven Lables

Month

All

Manufactured Quantity



110M

Rejected Quantity



491K

Processed Quantity



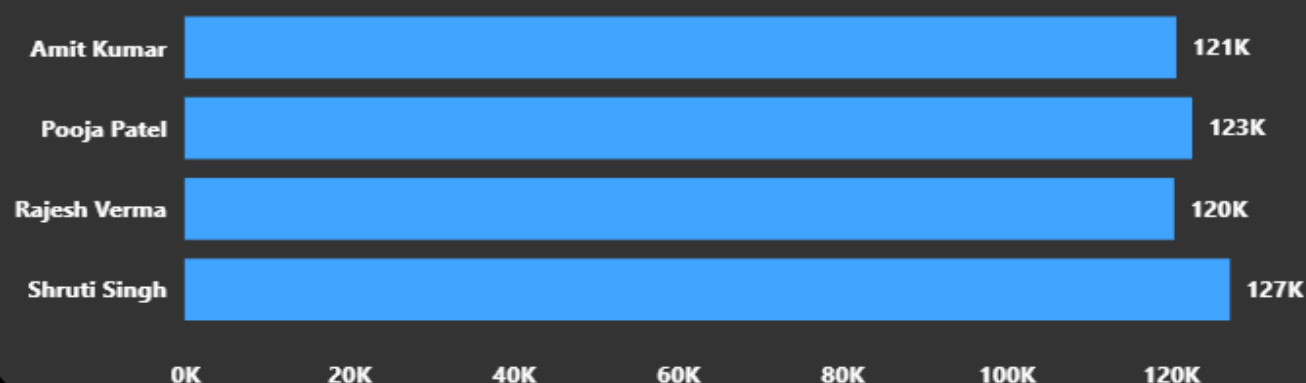
60M

Total Wastage %

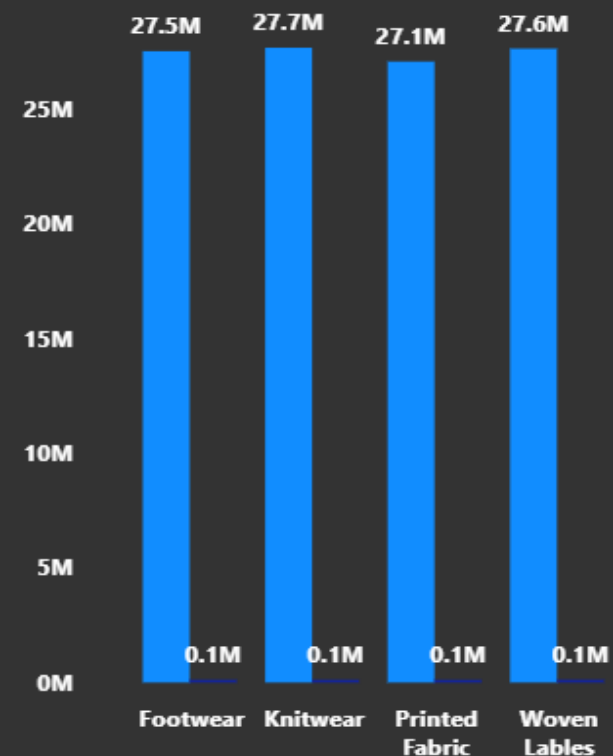


0.45

Employee Wise Rejected Qty



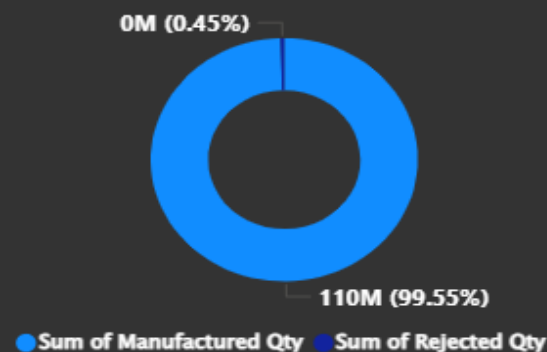
Department Wise Manufacture Vs Rejected



Production Comparison trend



Manufacture Vs Rejected





SQL QUERIES

1. Manufacture Qty:

Total quantity of items produced within a given timeframe.

```
select sum(`Total_Manufactured`) as Manufacture_Qty  
from manufacturing_data;
```

Result Grid	
	Manufacture_Qty
▶	109841221

2. Rejected Qty:

Number of units that failed quality checks and were rejected.

```
select sum(`Rejected Qty`) as Rejected_Qty  
from manufacturing_data;
```

Result Grid	
	Rejected_Qty
▶	491023

3. Processed Qty:

Quantity of items that completed specific manufacturing stages.

```
select sum(`Produced Qty`) as Produced_Qty  
from manufacturing_data;
```

Result Grid	
	Produced_Qty
▶	60050621

4. Total Wastage Qty %:

Materials or items lost during the production process.

```
select concat(round((sum(` Rejected Qty `) / ((sum(`produced qty`) +  
sum(` Total_Manufactured `)* 100,2), '%')  
as Total_Wastage_Qty  
from manufacturing_data;
```

Result Grid		Filter Rows:
	Total_Wastage_Qty	
▶	0.45%	



SQL QUERIES

5. Employee Wise Rejected Qty:

Rejection count attributed to individual employees for accountability and training.

```
select `Emp Name`, sum(`Rejected Qty`) as Emp_wise_rejected_qty  
from manufacturing_data  
group by `Emp Name`  
order by Emp_wise_rejected_qty desc;
```

	Emp Name	Emp_wise_rejected_qty
▶	Shruti Singh	127230
	Pooja Patel	122636
	Amit Kumar	120708
	Rajesh Verma	120449

6. Machine Wise Rejected Qty:

Tracks machine-specific rejections to identify underperforming equipment.

```
select `Machine Code`, sum(`Rejected Qty`) as Machine_wise_rejected_qty  
from manufacturing_data  
group by `Machine Code`  
order by Machine_wise_rejected_qty desc  
limit 5;
```

	Machine Code	Machine_wise_rejected_qty
▶	MC027	80049
	MC026	68459
	MC094	48897
	MC058	45918
	MC062	36687



SQL QUERIES

7. Production Comparison Trend:

Visual trends of production over time to analyze performance patterns.

```
select monthname(`Emp Name`) as Month_Name,  
SUM(`Produced Qty`) as Monthly_production_trend  
from manufacturing_data  
group by month(`Doc Date`), Month_Name  
order by month(`Doc Date`);
```

Result Grid	Filter Rows:
Month_Name	Monthly_production_trend
January	5075338
February	4597390
March	5399228
April	4855150
May	5003606
June	5251148
July	4921864
August	5094156
September	4717903
October	4800267
November	5136142
December	5198429

8. Manufacture Vs Rejected:

Direct comparison of produced and rejected quantities to assess quality ratios.

```
select (SUM(Total_Manufactured)) as Total_Manufactured,  
SUM(`Rejected Qty`) as Total_Rejected  
from manufacturing_data;
```

Result Grid	Filter Rows:
Total_Manufactured	Total_Rejected
109841221	491023



SQL QUERIES

9. Department Wise Manufacture Vs Rejected:

Department-level breakdown of production versus rejection to identify area-specific issues.

```
select `Department Name`, SUM(Total_Manufactured) as Manufactured_Qty,  
SUM(`Rejected Qty`) as Rejected_Qty,  
CONCAT(ROUND(SUM(`Rejected Qty`)*100/SUM(`Produced Qty`),2),"%") as Rej_Percent  
from manufacturing_data  
group by `Department Name`  
order by Rejected_Qty desc;
```

Result Grid Filter Rows: Export: Wrap Cell Content:				
	Department Name	Manufactured_Qty	Rejected_Qty	Rej_Percent
▶	Woven Lables	27618493	125795	0.84%
	Knitwear	27659115	122995	0.81%
	Footwear	27508729	122856	0.81%
	Printed Fabric	27054884	119377	0.81%

KEY TAKEAWAYS

BUSSINESS INSIGHTS

1. **Employee-wise rejections:** Highest from **Shruti Singh(127k)**, and lowest has **Rajesh Verma**.
2. **Operation-wise rejections:** Highest rejection in **Cut & Fold(118k)** and lowest rejection in **Laser cutting 2(145)**
3. **Machine-wise rejections:** Top 5 rejections – **MC027**, **MC026**, **MC094**, **MC058**, **MC062**
4. **Rejection Rate:** **0.45%**
5. **Reject Qty:** **4,91,023 (491k)**

CHALLENGES & FUTURE RECOMMENDATIONS

1. Schedule **regular preventive maintenance** for critical machines (**MC027, MC026, MC094, MC058, MC062**) to minimize downtime.
2. Enhance **quality inspections** at key stages, with special focus on the top 2 processes: **Cutting & Folding** and **Printing**.
3. Provide **employee training programs** aimed at minimizing operational mistakes and improving process efficiency.
4. Deploy a **real-time rejection monitoring dashboard** to enable early detection and quick corrective action.
5. Establish **KPI-based accountability** at employee, machine, and process levels to monitor and improve quality outcomes.
6. Drive **reject cost reduction initiatives**, targeting recovery of approximately **₹134,996 annually**.



THANK YOU